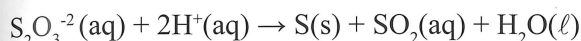


Experiment 40: Temperature and rate of reaction

Temperature can have a significant impact on the rate of reactions. We use refrigeration to reduce rates of spoiling of food. We heat other reactions to change their rate.

The purpose of this experiment is to investigate the effect of temperature on reaction rate. The reaction observed is the same as in the previous experiment, between sodium thiosulfate and hydrochloric acid.



The time taken for the precipitate of sulfur to obscure a cross marked on a piece of paper placed under the reaction flask is used as a measure of the reaction rate. The less time taken for the cross to 'disappear', the faster is the rate of reaction.

Equipment

hydrochloric acid [HCl] 2 mol L ⁻¹ (30 mL)	distilled water
conical flask (100 mL)	test tube
thermometer (-10 to 110 °C)	stop clock
Bunsen, tripod and gauze mat	paper towel
beaker (500 mL) to be used as a water bath	
sodium thiosulfate solution [Na ₂ S ₂ O ₃] 0.25 mol L ⁻¹ (150 mL)	
graduated cylinders (10 mL, 25 mL and 100 mL)	



Notes

SAFETY NOTE:

- Avoid breathing sulfur dioxide evolved while waiting for the dot to disappear. If asthmatic carry out in a fume cupboard.

Procedure

- Place 15 mL of 0.25 mol L⁻¹ Na₂S₂O₃ and 30 mL of water into a 100 mL conical flask.
- Measure 5 mL of 2 mol L⁻¹ HCl into a test tube.
- Heat the solutions separately in a water bath until they are slightly above 20 °C. Remove the flask from the water bath and dry the outside of the flask with paper towel.
- Put the flask over a cross marked on a piece of white paper and add the acid to the thiosulfate solution. Start a stop clock at the moment of addition.
- Note the time taken for the cross to 'disappear' and record the average temperature during the reaction.
- Repeat the procedure for temperatures of about 30, 40, 50 and 60 °C. In each case you will need to measure the average temperature for the reaction.
- Construct a table showing the temperature, times taken for the cross to disappear, and 1/time.

Processing of results and questions

- Plot a graph of 1/time against temperature.
- What effect does increasing the temperature have on the reaction rate?
- Suggest two factors that contribute to the change in reaction rate with temperature.
- Identify two random errors in this experiment. How could these errors be minimised?